

Audit and simplified truss layouts

The W4-N1 upper bay had TWO superposed systems: narrow original truss panels plus a later broad X-brace. It now uses one diagonal spanning the whole bay. TE also had an X-brace superposed on intermediate uprights; it now uses one upper-bay diagonal. Old roof-cap layout uprights that crossed webs were removed.

17 uprights/hangers were merged into existing columns or side-truss uprights. 12 other members were removed. 30 redundant localized joint links were coalesced. TS end uprights use W1/S3. T1 end uprights and end hangers use TE/TW uprights. Shared member stiffness and self-weight occur once in the global model.

Retained deliberately:

- T1 upper truss and lower floor/trolley rail: different elevations and functions.
- TW balcony jamb at $y=247$ in: 6 in from W4; preserves the door opening.
- Ground and roof-plan X-braces: lateral systems, not duplicate upper truss webs. Their crossings/clearances and gusset details remain to be designed. Roof-plan braces still pass the T1 chord; crossing hardware is not resolved here.

No ground columns, floor area, roof envelope, solar allowance or occupancy loads were removed. Section sizes are unchanged; roof-first and final stages were rerun.

Steel model: 9,322 → 8,512 lb (810 lb removed).

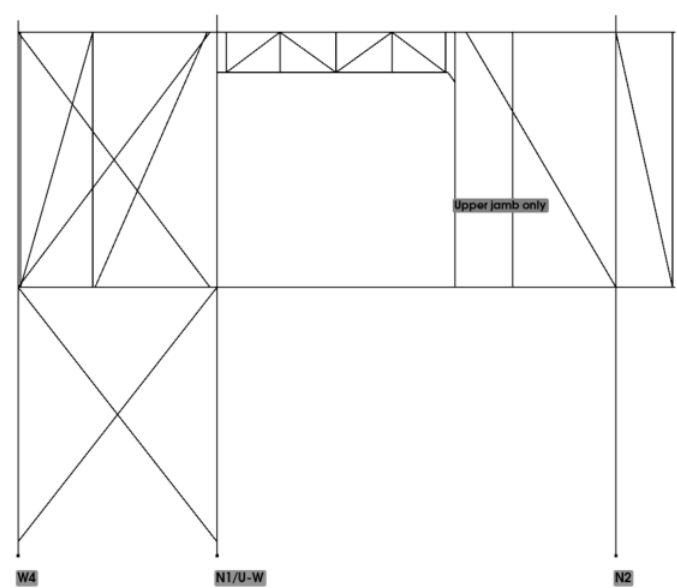
Partial second-order screen: 0.834 → 0.935; W3 governs.

Preliminary model. Not a complete strength, deflection, connection, foundation or code approval.

TN Full wall elevation

BEFORE — overlapping upper bracing

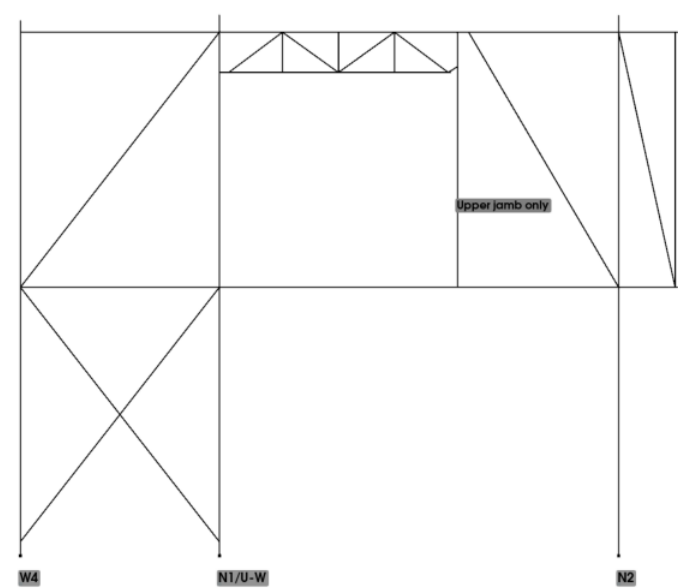
Flat upper chord axis: 18 ft 8 1/4 in
Lower chord axis: 9 ft 7 in
Overall chord run: 23 ft 5 1/2 in



TN Simplified wall elevation

AFTER — single upper-bay diagonal

Flat upper chord axis: 18 ft 8 1/4 in
Lower chord axis: 9 ft 7 in
Overall chord run: 23 ft 5 1/2 in

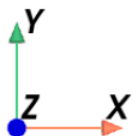
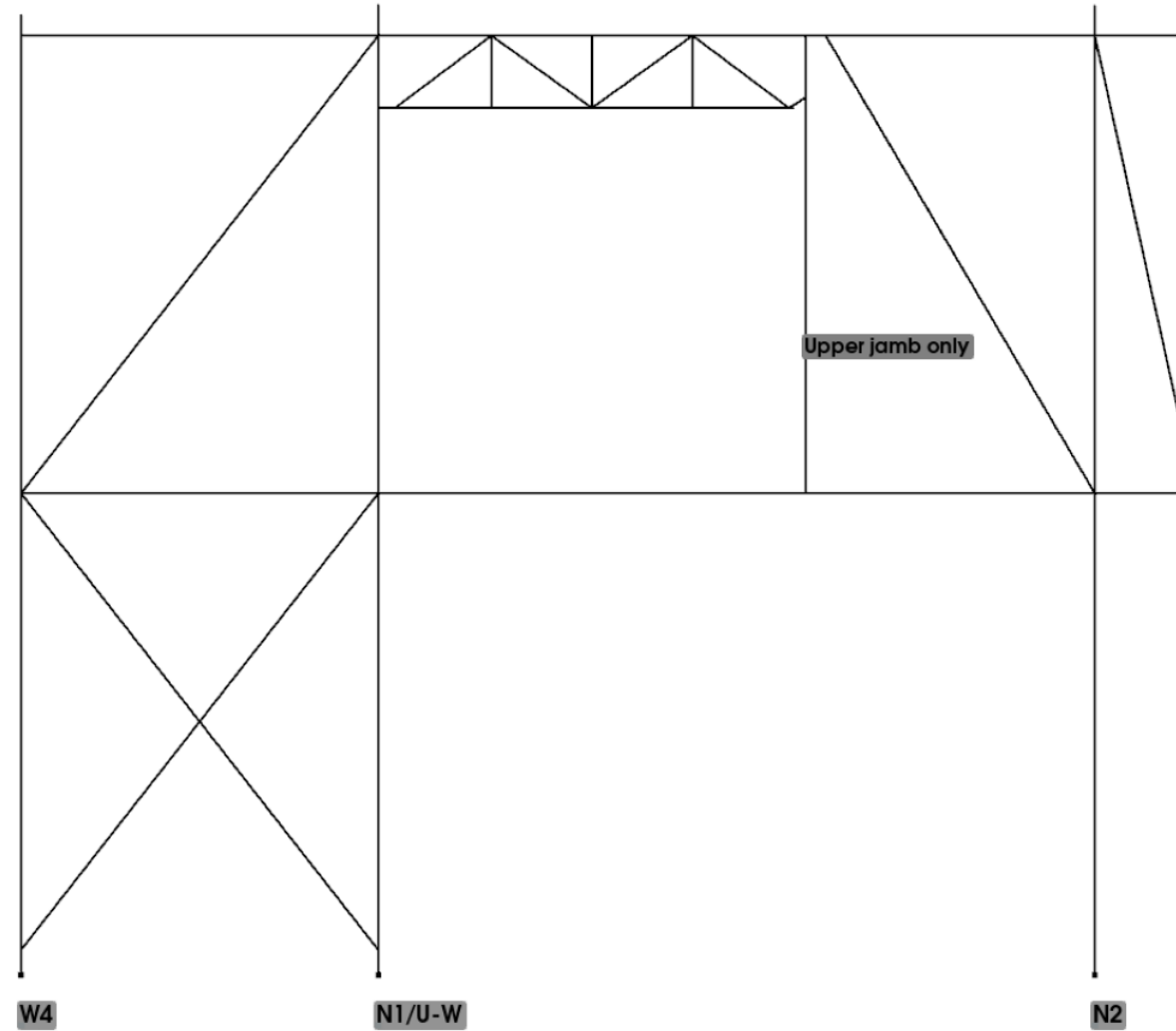


Actual ground supports shown. Shared columns occur once in the global model.
Fixed-base analytical assumption; foundations and connections not designed.

Shared columns replace duplicate truss uprights. Ground-level X bracing retained.
Preliminary fixed-base model; connection and foundation design remains.

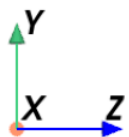
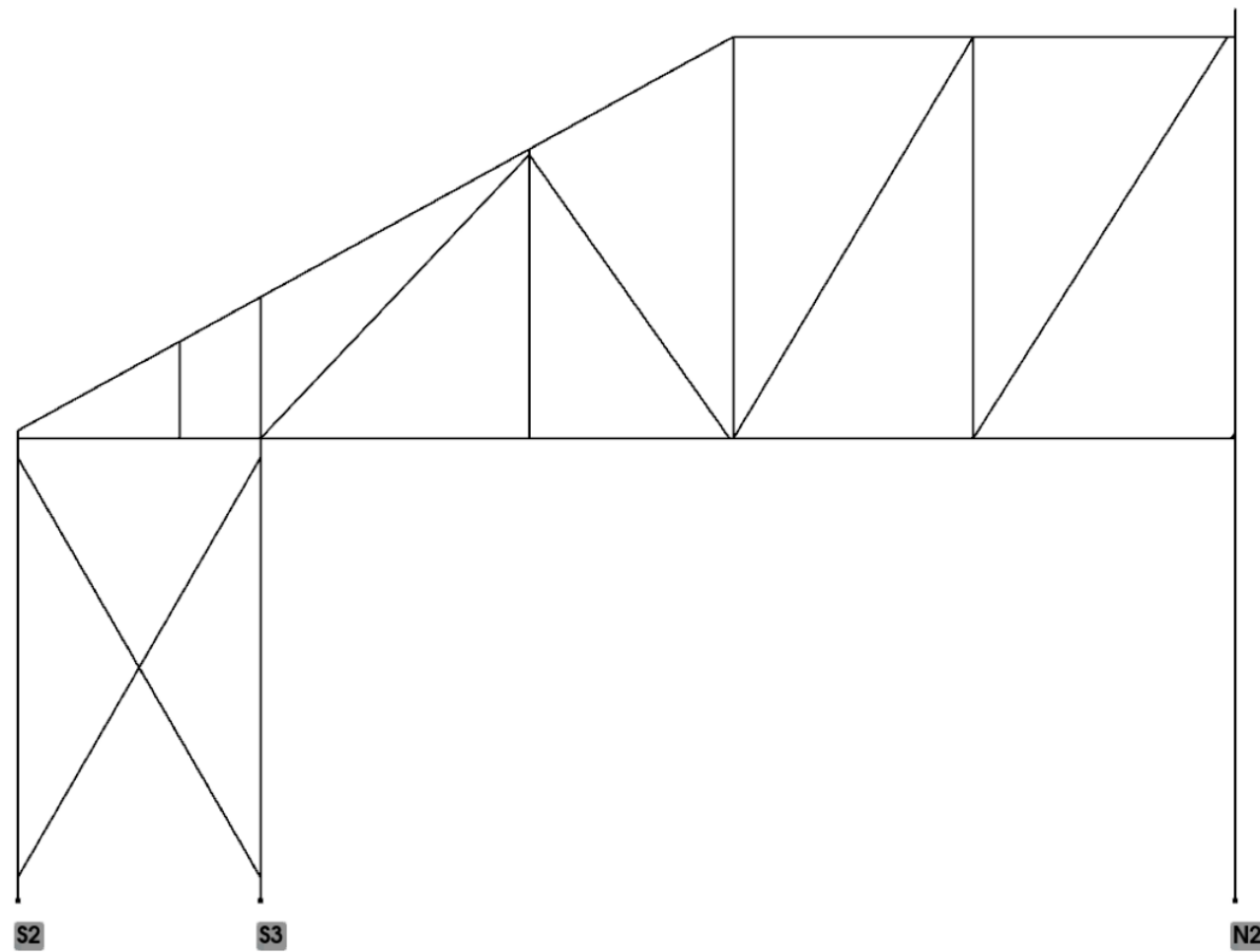
TN Simplified wall elevation

Flat upper chord axis: 18 ft 8 1/4 in
Lower chord axis: 9 ft 7 in
Overall chord run: 23 ft 5 1/2 in



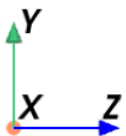
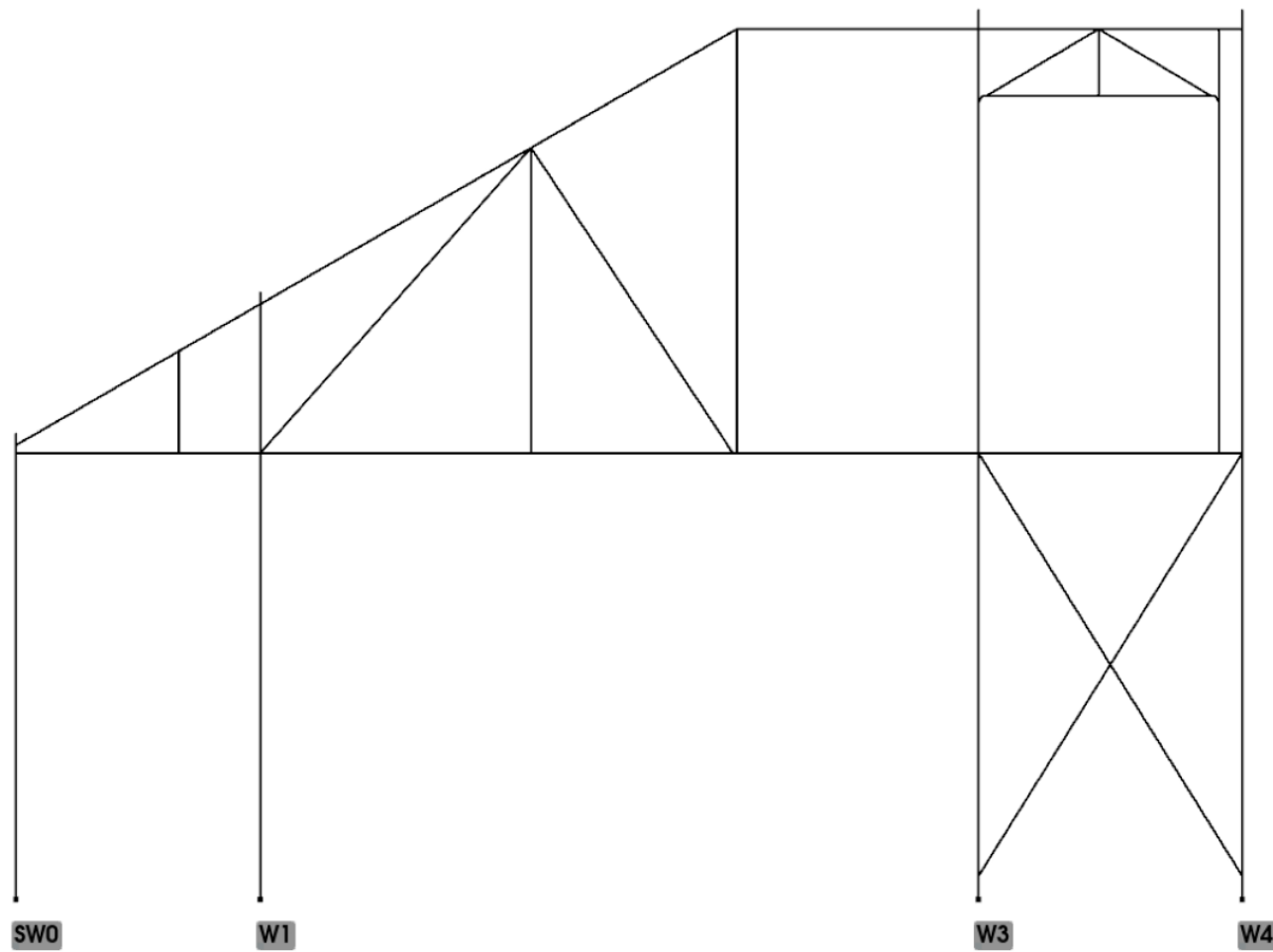
Shared columns replace duplicate truss uprights. Ground-level X bracing retained.
Preliminary fixed-base model; connection and foundation design remains.

TE Simplified wall elevation



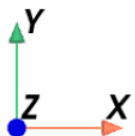
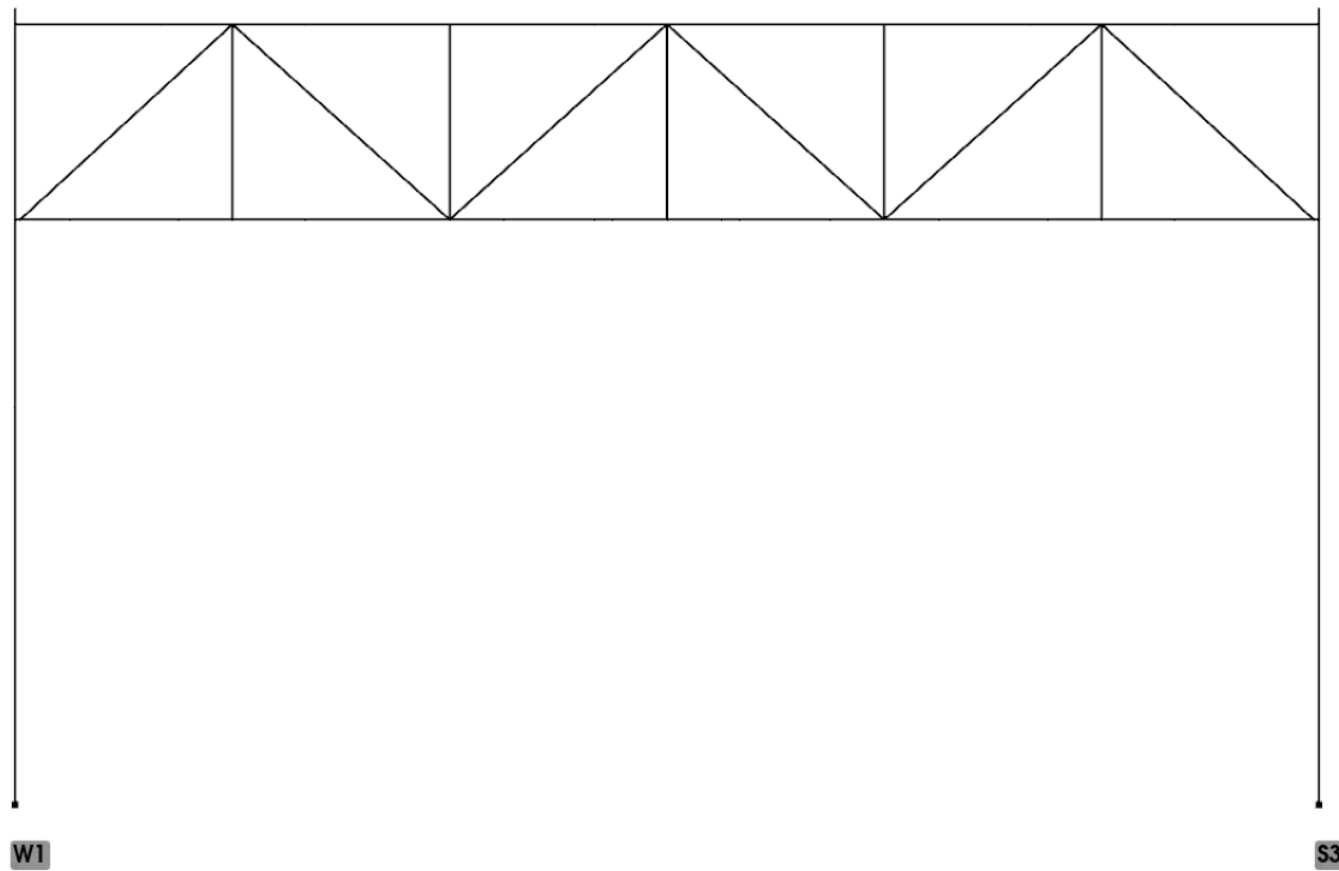
Shared columns replace duplicate truss uprights. Ground-level X bracing retained.
Preliminary fixed-base model; connection and foundation design remains.

TW Simplified wall elevation



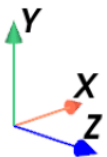
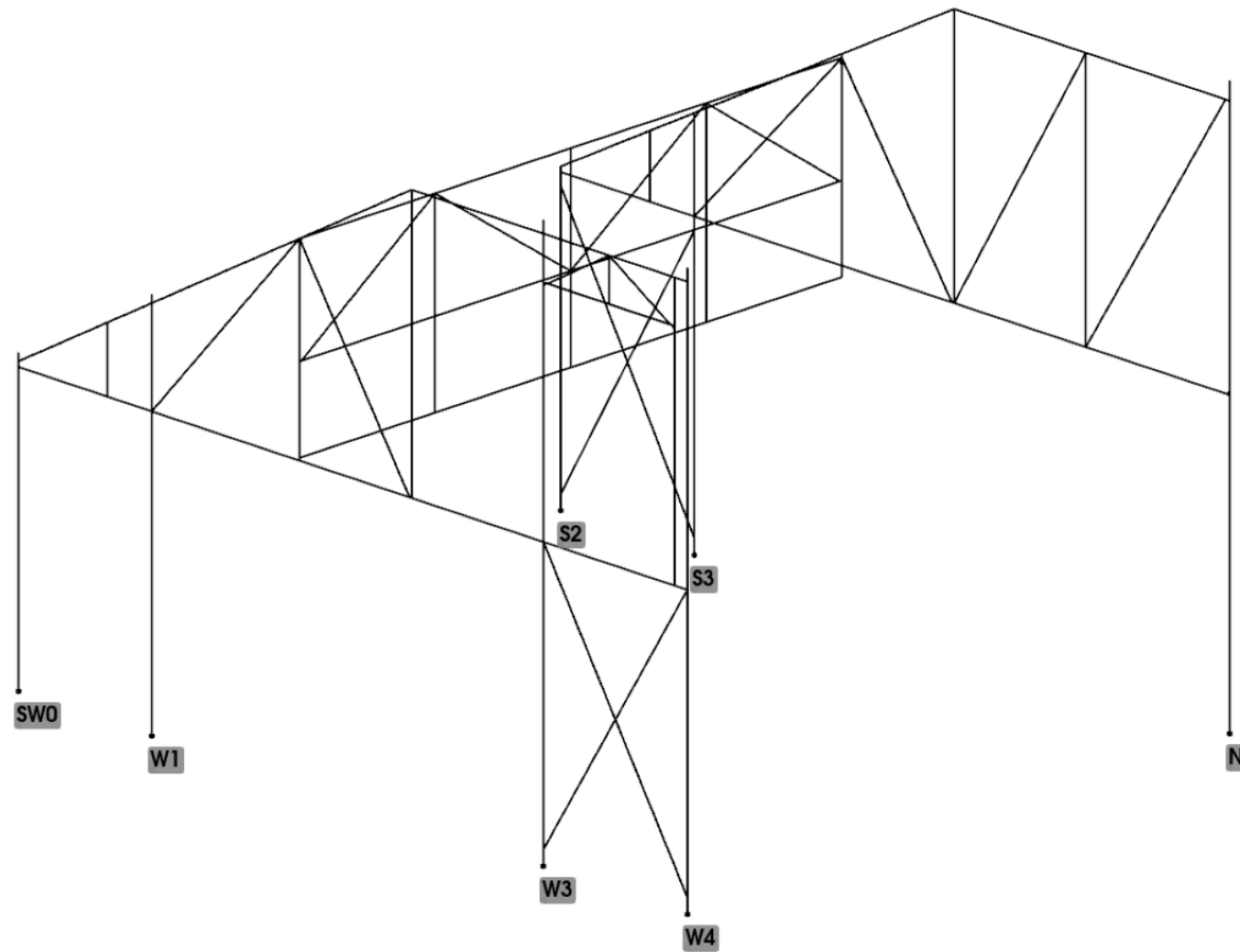
Shared columns replace duplicate truss uprights. Ground-level X bracing retained.
Preliminary fixed-base model; connection and foundation design remains.

TS Simplified wall elevation



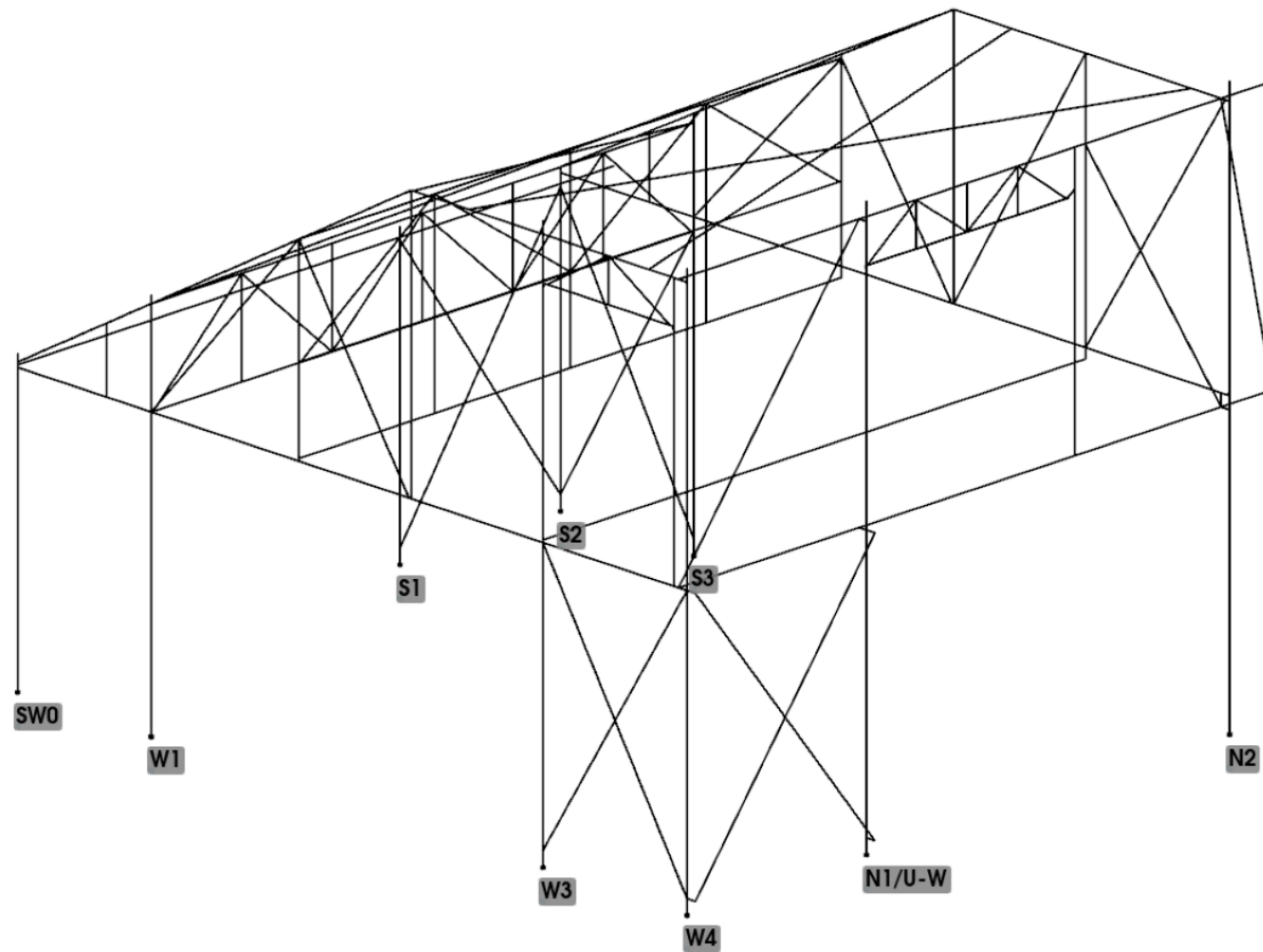
Shared columns replace duplicate truss uprights. Ground-level X bracing retained.
Preliminary fixed-base model; connection and foundation design remains.

T1 Simplified support path



T1 bears on TE and TW; those trusses transfer its load to the ground columns.
No new posts beneath T1. This is a filtered 3D view of the globally solved frame.

ALL Simplified support path



Shared columns replace duplicate truss uprights. Ground-level X bracing retained.
Preliminary fixed-base model; connection and foundation design remains.

Member-by-member audit disposition

T-S vertical 1 → W1
T-S vertical 7 → S3
T-W vertical 2 → W1
T-W vertical 5 → W3
T-W vertical 7 → W4
T-W vertical 11 → W3
T-W vertical 14 → T-W vertical 6
T-E vertical 2 → S3
T-E vertical 9 → N2
T-N vertical 1 → W4
T-N vertical 4 → N2
T-N vertical 10 → N1 / U-W
T-N vertical 14 → T-N upper east jamb
T1 vertical 1 → T-W vertical 3
T1 vertical 5 → T-E vertical 3
T1 future hanger 1 → T-W vertical 3
T1 future hanger 5 → T-E vertical 3

Removed:

BR-E-upper-2
BR-N-upper-2
T-E diagonal 10
T-E vertical 5
T-E vertical 7
T-E vertical 8
T-N diagonal 6
T-N diagonal 7
T-N vertical 2
T-N vertical 3
T-W diagonal 8
T-W vertical 12